Decoder-Free Optical Learning for Axial Structure Recovery in Fixed-Focus Fluorescence Microscopy

Supplementary Material

Sections S1 to S5 specify the optical model, training, reconstruction and evaluation. Full results, structural tests and acquisition controls follow in Secs. S6 to S8; Secs. S9 and S10 give complementary uncertainty analyses and neuronal display conventions. Section, equation and figure numbers without an S prefix refer to the main paper. Unless stated otherwise, additional experiments use the 12 prespecified neuronal crops, with five direction subsets at $C > 1$.

S1. Optical implementation and discretization

Coordinates and angular scan. The optical axis is z and $\mathbf{r} = (x, y)$ is transverse position. In the simulated input medium, $n = 1$ and $k_{\mathrm{ex}} = 2\pi/\lambda_{\mathrm{ex}}$. The incident direction is

$$\mathbf{k}_t/k_{\mathrm{ex}} = (\cos\theta_t \cos\alpha_t, \cos\theta_t \sin\alpha_t, \sin\theta_t). \tag{S1}$$

Thus, θ is the implemented vertical-plane parameter, not a polar angle from the optical axis. We use $\alpha_m = m\pi/16$, $m = 0, \ldots, 15$, with eight equally spaced θ values in each of $[\pi/6, \pi/3]$ and $[2\pi/3, 5\pi/6]$. The second interval changes the sign of the transverse component while retaining positive longitudinal propagation. The resulting 256 directions are distinct. For $C = 4$ and 8, five fixed random subsets of 64 and 32 directions are shared across encoders.

Propagation and sampling. The incident field is uniformly x-polarized before phase modulation. Angular-spectrum propagation over $D = 6.6\,\mu\mathrm{m}$ carries the field from the plate to the intermediate illumination image plane. A finite NA-0.8 pupil filters the modulated angular spectrum before a vectorial Richards–Wolf relay generates three specimen-field components [7, 9]. Incident direction and transmitted angular spectrum are different quantities; the input direction range need not lie entirely inside the relay pupil. We sum the component intensities. Detection uses an NA-0.1 emission intensity PSF. Excitation and emission share 532 nm in this simulation; a physical implementation would use the actual spectral bands. No scattering, aberration or fluorescence saturation is modeled.

The specimen field is sampled laterally at $0.08\,\mu\mathrm{m}$ and axially at $0.32\,\mu\mathrm{m}$. Lateral binning gives an isotropic $0.32\,\mu\mathrm{m}$ analysis grid of $32 \times 96 \times 96$ cells. The reconstruction basis averages $2 \times 2 \times 2$ analysis cells into each $0.64\,\mu\mathrm{m}$ voxel. Camera pixels average 8×8 analysis samples laterally, giving $2.56\,\mu\mathrm{m}$ pixel pitch and 12×12 samples per image. Distances and pitches are expressed in specimen-equivalent coordinates.

Image formation. Let $X_0(\mathbf{r}, z)$ denote fluorescence response per unit excitation and h_{em} the emission intensity PSF. With linear fluorescence and laterally shift-invariant detection, the noiseless camera-plane irradiance is

$$g_t(\mathbf{r}) = \int_\Omega h_{\mathrm{em}}(\mathbf{r} - \mathbf{r}', z_{\mathrm{focal}} - z') \times X_0(\mathbf{r}', z') I_t(\mathbf{r}', z')\, \mathrm{d}^2\mathbf{r}'\, \mathrm{d}z'. \tag{S2}$$

Camera pixel p, occupying D_p, records $\bar{y}_{t,p} = |D_p|^{-1} \int_{D_p} g_t(\mathbf{r})\, \mathrm{d}^2\mathbf{r}$. Emission adds incoherently over the specimen before this average. Excitation multiplies the specimen before detection blur; the detector need not resolve the illumination pattern itself.

Aperture and control encoders. The phase plate has $0.32\,\mu\mathrm{m}$ features and initially independent phases uniform on $[0, 2\pi)$. It is simulated on an aperture twice the specimen width, then the excitation is cropped to the specimen. Only the central plate region is optimized; its independent random surround remains fixed. The defocused amplitude controls use binary masks with the same feature pitch and 50 % open pixels at the same plate plane. The in-focus DMD uses independently drawn binary patterns relayed into focus. These are controls for the tested fixed-focus acquisition, not a comparison against all DMD microscopy configurations.

A finite illumination edge can itself encode depth when tilted. The plate-free control therefore uses an aperture four times the specimen width. Its residual central/border angular modulation is about 1.0/1.2 %, compared with 5.6/37 % at twice the width. The $4\times$-aperture plate-free study is kept separate from the $2\times$ phase/mask comparison. The random plate changes little across these apertures (centroid error 0.446 versus 0.431 μm at $C = 8$, pretrained PnP), but that does not make their learned-versus-plate-free contrast a single-aperture experiment.

Integrated operator. The detection footprint $g_{p,i}$ includes the emission PSF, camera-pixel average and fine-cell integration weight. Reusing it in $A_{(t,p),j} = \sum_{i \in \mathcal{P}_j} I_t(\mathbf{r}_i, z_i) g_{p,i}$ avoids propagating each reconstruction basis element separately. Equality $\mathsf{A} = \mathsf{BU}$ is an algebraic statement for the selected piecewise-constant basis. It does not assert that a real specimen is constant within a voxel. For a fine-grid specimen $\mathbf{x}_f$, the voxel-averaged reference is $\mathbf{x}_{\mathrm{gt}} = \mathsf{R}\mathbf{x}_f$, where R averages within reconstruction voxels. Normalized

noisy measurements satisfy

$$\mathbf{y} = \mathsf{A}\mathbf{x}_{\mathrm{gt}} + \underbrace{\mathsf{B}(\mathbf{x}_f - \mathsf{U}\mathsf{R}\mathbf{x}_f)}_{\boldsymbol{\delta}_{\mathrm{disc}}} + \boldsymbol{\varepsilon}. \tag{S3}$$

The retained residual is caused by specimen structure outside the reconstruction basis, not by an algebraic discrepancy between $\mathsf{A}\mathbf{x}$ and $\mathsf{B}\mathsf{U}\mathbf{x}$.

Computational scope. After detection-footprint precomputation, direct assembly costs $O(TPF)$, compared with $O(TNF\log F)$ for constructing columns through separate FFT-based forward simulations: T illumination patterns, P camera samples, F fine-grid cells and N reconstruction voxels. Representative timings on an RTX 4500 Ada are approximately 20 minutes for 1000 optical-training steps, six minutes to assemble a full plate operator, 4–6 seconds for a PnP reconstruction at $C = 8$, and 17 seconds for TV.

S2. Plate training and objective comparisons

Signatures and sampling. For each lateral position and depth, we collect the detector-integrated response of its reconstruction voxel over the sampled illumination directions and camera pixels. Four directions and 24 interior lateral positions are sampled per Adam step. Each full-field depth signature therefore has 4×144 entries during training. Its mean over all angle–pixel entries is removed before unit-norm normalization. This is not per-pixel temporal centering. The main paper's evaluation statistic uses the same centering definition over the 32 directions of draw 0 and the 44×44 interior lateral positions (indices 2–45). It averages absolute off-diagonal depth correlations, not the squared weighted training loss.

With zero-based reconstruction-plane indices $i, j \in \{0, \ldots, 15\}$, off-diagonal weights are

$$w_{ij} = \begin{cases} 2, & i + j = 14, \\ 1, & i + j \neq 14, \end{cases} \qquad i \neq j. \tag{S4}$$

This emphasizes reflected pairs around grid plane 7. It is an implemented weighting choice, not a universal symmetry of a phase screen or a statement that plane 7 is exactly the physical focus. Both orderings of each pair are included and the sum is divided by the total weight. Norm denominators include 10^{-12} for numerical stability.

We use 1000 Adam steps with learning rate 0.05. The two training seeds change the sampled positions and directions, starting from the same phase realization and frozen surround. No specimen examples, reconstruction loss or denoiser enter this process. Normalizing a signature removes uniform power scaling from the loss; dose normalization is applied separately when generating evaluation measurements.

Local Gaussian interpretation. At one lateral position, let Q collect Z unit-norm depth signatures and $\mathsf{G} = \mathsf{Q}^\top\mathsf{Q}$. Then $\operatorname{tr}\mathsf{G} = Z$ and the unweighted off-diagonal sum is $\|\mathsf{G} - \mathsf{I}\|_F^2$. Consider the idealized observation $\mathbf{y} = \mathsf{Q}\mathbf{x} + \boldsymbol{\epsilon}$, with independent $\mathbf{x} \sim \mathcal{N}(0, \sigma_x^2\mathsf{I})$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_e^2\mathsf{I})$. Writing $\gamma = \sigma_x^2/\sigma_e^2$, the mutual information and linear-MMSE are

$$I(\mathbf{x};\mathbf{y}) = \tfrac{1}{2}\log\det(\mathsf{I} + \gamma\mathsf{G}), \tag{S5}$$

$$\mathrm{MSE}_{\mathrm{lin}} = \sigma_x^2 \operatorname{tr}(\mathsf{I} + \gamma\mathsf{G})^{-1}. \tag{S6}$$

At fixed trace, concavity of the log and convexity of the inverse favor equal eigenvalues, giving the orthogonal optimum $\mathsf{G} = \mathsf{I}$ when attainable. Set $\mathsf{E} = \mathsf{G} - \mathsf{I}$. Since $\operatorname{tr}\mathsf{E} = 0$, the expansions near this point are

$$I = \tfrac{Z}{2}\log(1+\gamma) - \tfrac{1}{4}\left(\frac{\gamma}{1+\gamma}\right)^2 \|\mathsf{E}\|_F^2 + O(\|\mathsf{E}\|_F^3), \tag{S7}$$

$$\mathrm{MSE}_{\mathrm{lin}} = \sigma_x^2\left[\frac{Z}{1+\gamma} + \frac{\gamma^2}{(1+\gamma)^3}\|\mathsf{E}\|_F^2 + O(\|\mathsf{E}\|_F^3)\right]. \tag{S8}$$

This motivates decorrelation of axial responses. It does not derive the implemented weights, centering or normalization from the full Poisson specimen model. The actual learned signatures remain correlated, other lateral columns interact, and the full reconstruction is not a Gaussian linear-MMSE estimator. The calculation is a local interpretation of a known Gram-matrix design principle [3, 4], not a transfer guarantee or an error bound for neuronal W1.

Why angular modulation matters. Consider a nonnegative temporal response $\mathbf{e}_z$ at one camera pixel, with mean $\mu_z > 0$. Write $\mathbf{e}_z = \mu_z\mathbf{1} + \tilde{\mathbf{e}}_z$, and define angular contrast $\kappa_z = \|\tilde{\mathbf{e}}_z\|/(\sqrt{T}\mu_z)$. If ρ is the correlation of the mean-removed fluctuations, the raw cosine is exactly

$$c_{zz'} = \frac{1 + \rho\kappa_z\kappa_{z'}}{\sqrt{(1+\kappa_z^2)(1+\kappa_{z'}^2)}}. \tag{S9}$$

The shared mean dominates when contrasts are small, even if the fluctuations are weakly correlated. Increasing contrast and placing strong responses at different angles can reduce this overlap. At the preset field center, median contrast is 0.58/1.53 and mean raw cosine 0.83/0.41 for random/learned plates. Only one preset position is used for this illustrative decomposition; it is not a field-wide causal ablation.

A common-footprint approximation explains why centering the full angle–pixel signature does not necessarily remove the shared angular component. If $\mathbf{q}_z = \mathbf{e}_z \otimes \mathbf{g}$ with a fixed P-pixel footprint, global centering gives correlation

$$\frac{1 - \eta + \rho\kappa_z\kappa_{z'}}{\sqrt{(1-\eta+\kappa_z^2)(1-\eta+\kappa_{z'}^2)}}, \qquad \eta = \frac{(\mathbf{1}^\top\mathbf{g})^2}{P\|\mathbf{g}\|^2}. \tag{S10}$$

For a localized footprint, η can be small, so a common angular component remains. The actual operator includes depth-varying footprints and within-voxel variation; neither training nor evaluation uses this separable approximation. This provides an interpretation of the observed code, not a proof that contrast alone causes the reconstruction gain or that the objective is universally optimal.

Worst-case coherence surrogate. Let ρ_{ij} be the inner product of two centered, unit-norm depth signatures. The comparison uses

$$\mathcal{L}_{\mathrm{coh}} = \mathbb{E}_{\mathbf{r},\mathcal{T}}\left[\frac{1}{50}\log\sum_{i\neq j}\exp(50|\rho_{ij}|)\right]. \tag{S11}$$

This smooth maximum targets the worst pair rather than the weighted average of squared correlations. It is motivated by coherence-based sensing and PSF design [3, 4, 13]; it does not implement Miniscope3D's full optical design or lateral-shift cross-coherence objective.

Depth Fisher-information surrogate. Let R contain the raw signatures in a sampled batch. The implementation uses $s = R/\operatorname{stopgrad}(\max(\operatorname{mean} R, 10^{-30})) + 10^{-3}$. Finite differences s'_z are taken over depth-plane index, centrally inside the grid and one-sided at its ends. With brightness treated as a nuisance parameter, the depth-information surrogate is

$$\mathcal{I}_z = \sum_m \frac{(s'_{z,m})^2}{s_{z,m}} - \frac{(\sum_m s'_{z,m})^2}{\sum_m s_{z,m}}, \tag{S12}$$

where m indexes angle–pixel entries. Training minimizes $-\operatorname{mean}_z \log\max(\mathcal{I}_z, 10^{-12})$. This uses the Schur-complement form of Poisson Fisher information [10], with the stated normalization and discretization. It is an implemented surrogate, not a comparison with every Fisher-based design.

Matched budget and centering ablation. All objectives use the same initial phase, sampled positions/directions, optimizer, learning rate and step count. Reconstruction parameters are validated independently for each resulting operator. These conditions match computational budgets, not convergence rates or optimal hyperparameters for every criterion. The uncentered variant omits mean removal and retains the weights and squared-correlation form. On the 12 prespecified crops with neuron-trained PnP at $C = 8$, it is essentially tied with the centered version (W1 0.207 versus $0.206\,\mu$m). Centering is therefore not an established explanation of the gain over the tested alternatives.

Complete 60-crop objective comparison. The primary five-design table uses all 60 test crops and five draws per crop. The two alternative criteria are evaluated with TV, pre-trained PnP, neuron-trained PnP and separately fine-tuned feed-forward decoders. Table S1 reports their full metrics, and Tab. S2 gives paired W1 changes relative to our sampling-seed-0 plate. The corresponding untrained, our and end-to-end rows are in Tab. S4. These tables use complete configurations only. Table S3 repeats the TV/PnP comparison with sampling seed 1 on the same 60 crops. Each criterion is compared with our matching sampling seed; all paired W1 intervals exclude zero.

S3. Joint optical–decoder training

Shared optics. The jointly trained counterpart uses the same phase parameterization, initial phase, frozen random surround, vectorial relay, detector footprints and reconstruction basis as the model-only designs. The measurement matrix is assembled differentiably from the excitation and fixed detection footprints. During joint training, each sampled 32-direction excitation is normalized by its own specimen-field mean. Evaluation instead uses the full-scan dose normalization shared by every encoder. This difference is retained explicitly in the training protocol; model-only correlation losses are invariant to a uniform scale.

Regularized solve and network. For $M = 32 \times 144 = 4608$ measurements and $N = 36{,}864$ coefficients, the decoder first computes

$$\mathbf{z} = \mathbf{m} + \mathsf{A}^\top(\mathsf{A}\mathsf{A}^\top + \lambda\mathsf{I})^{-1}(\mathbf{y} - \mathsf{A}\mathbf{m}). \tag{S13}$$

Here $\mathbf{m}$ is a constant volume filled with the training set's mean intensity. A learned log-parameter sets $\lambda > 0$, initialized from mean training noise variance divided by training voxel variance. The solve is performed in measurement space. The Gram matrix is formed in float32 with TF32 disabled, and its Cholesky factorization and triangular solve use float64. The result is projected back in float32. One such solve is followed by one residual 3D U-Net.

The network has three encoder widths, 16, 32 and 64. Each block contains two $3 \times 3 \times 3$ convolutions with batch normalization and ReLU. Two $2 \times 2 \times 2$ max-pooling stages reduce the spatial grid. Transposed convolutions of kernel and stride two upsample; lateral skip features are concatenated before the corresponding convolution blocks. A final $1 \times 1 \times 1$ convolution predicts a residual that is subtracted from the network input. The output has no nonnegative or box activation.

Training data and noise. Training uses the same 459 neuronal crops from stacks 8 and 9 of the fourth animal as the denoiser study. Stack 10 contributes 240 held-out crops,

Reconstruction	Plate	W1 (μm)	Centroid (μm)	PSNR	FG-SSIM
TV	Depth Fisher	0.251	0.175	38.33	0.908
TV	Worst-case coherence	0.264	0.185	37.99	0.898
Pretrained PnP	Depth Fisher	0.367	0.298	37.09	0.882
Pretrained PnP	Worst-case coherence	0.390	0.320	36.85	0.871
Neuron-trained PnP	Depth Fisher	0.218	0.157	39.35	0.927
Neuron-trained PnP	Worst-case coherence	0.223	0.160	39.19	0.924
Feed-forward	Depth Fisher	0.314	0.234	37.40	0.896
Feed-forward	Worst-case coherence	0.314	0.238	37.31	0.892

Table S1. Alternative model-only objectives on all 60 crops, $C = 8$, sampling seed 0. Each row averages five draws per crop and uses the same reconstruction for all four metrics. W1 and centroid errors are in μm.

Reconstruction	Ours vs. depth Fisher	Ours vs. worst-case coherence
TV	-13.1% [-14.4, -11.5]	-17.5% [-18.8, -16.3]
Pretrained PnP	-20.9% [-22.8, -18.9]	-25.7% [-27.8, -23.4]
Neuron-trained PnP	-10.2% [-10.8, -9.3]	-12.3% [-12.8, -11.6]
Feed-forward	-15.4% [-17.0, -14.0]	-15.4% [-16.8, -14.2]

Table S2. Relative W1 changes and 95 % cell-bootstrap intervals for the primary 60-crop criterion comparison. Negative changes favor our plate. Each criterion shares the initial phase and model-only optimization budget.

Comparator	Reconstruction	Other W1	Our W1	Change [95% CI]
Depth Fisher	TV	0.253	0.218	-13.9% [-15.7, -12.2]
Depth Fisher	Pretrained PnP	0.370	0.289	-22.0% [-24.8, -19.2]
Depth Fisher	Neuron-trained PnP	0.219	0.194	-11.5% [-12.9, -10.1]
Worst-case coherence	TV	0.263	0.218	-17.1% [-18.1, -16.0]
Worst-case coherence	Pretrained PnP	0.378	0.289	-23.7% [-25.9, -21.4]
Worst-case coherence	Neuron-trained PnP	0.220	0.194	-11.9% [-12.7, -10.8]

Table S3. Objective comparison with the second optical sampling seed: 60 crops, $C = 8$, five draws per crop. W1 is in μm; relative changes compare our plate with the comparator of the same sampling seed, with paired cell-bootstrap 95 % intervals. Both seeds start from the same initial phase. Negative changes favor our criterion.

of which every fifth crop gives 48 validation volumes. Each validation uses two fixed direction subsets and fixed noise, shared by all optical designs. Test animals never enter these stages. Training augments volumes with independent flips along each axis and lateral transposition.

Supervised training generates measurements from the coarse reconstruction-basis model $\bar{\mathbf{y}} = \mathbf{A}\mathbf{x}$. The noise approximation is

$$\widetilde{y}_i = \bar{y}_i + \sqrt{\max(\bar{y}_i, 0)/\kappa_k + 10^{-12}}\,\xi_i, \quad \xi_i \sim \mathcal{N}(0,1), \tag{S14}$$

with $\kappa_k = \kappa_0\, 256/32$ and per-crop exposure calibrated using the untrained plate. Test measurements use the common fine-grid specimen model and exact Poisson sampling. This preserves the sub-voxel specimen residual during evaluation rather than adding it to supervised training targets.

Optimization schedule. All supervised stages use batch size 16 and reconstruction MSE. The initial decoder is warmed up for 5000 Adam steps on the frozen untrained plate at learning rate 10^{-3}. Plate and decoder are then updated jointly for 1200 plate steps. The plate learning rate starts at 0.05 and follows a cosine decay to 0.005 over the last 30 % of steps. Decoder learning rate is 3×10^{-4}; each joint step is followed by four decoder-only steps on fresh batches using the same direction subset and a detached operator. The phase outside the trained central region remains fixed.

The initial 1200-step run did not meet a prespecified plateau criterion: validation MSE changed by 3.2 % over its last 300 plate updates, above the 2 % threshold. Before test evaluation, a fixed continuation of 600 updates was specified, with phase learning rate 0.005, unchanged decoder recipe, fresh optimizer states and restored model parameters.

The continuation's last-300-step change was 0.025 %. The selected phase checkpoint, at update 1500, minimizes validation MSE across the continuation checkpoints. Phase RMS change from initialization is 1.44 radians. These checks limit selection and verify training behavior; they are not a claim of global convergence.

Numerical checks. At initialization, differentiable assembly reproduces stored operator rows to relative error 8.5×10^{-8} after accounting for the subset/full-scan normalization scalar. An adjoint check and three central finite-difference gradient projections pass; directional derivatives agree to 0.03–0.07 %. A short descent check confirms that the phase changes and the loss decreases. Cross-server reconstruction checks compare recorded image-quality and centroid metrics rather than requiring bit-identical floating-point builds. These checks precede transfer comparisons.

Decoder comparison after optical training. The phase is fixed before evaluation with TV, pretrained PnP or neuron-trained PnP. Each operator uses its validation-selected scalar parameters, while denoisers remain unchanged. For the feed-forward comparison, every plate receives 3000 additional decoder-only steps at learning rate 3×10^{-4}, with validation every 500 steps. The untrained and model-only plates start this stage from the same warmed decoder; the end-to-end plate starts from its selected jointly trained checkpoint. Thus the architecture and additional fine-tuning budget match, but the preceding decoder training histories do not. The model with lowest validation MSE is used for each plate.

Feed-forward outputs are retained without final clipping for PSNR, foreground SSIM and the implemented centroid metric. Axial-profile W1 uses their nonnegative part, as specified in Sec. S5. This differs from the final clipping of TV/PnP outputs and is shared across all plates within the feed-forward comparison.

Comparison scope and timing. The primary end-to-end comparison uses one optical initialization and one supervised training run. Its plate and decoder were optimized for MSE; the primary reported axial metric is W1. On the RTX 5080, decoder warm-up, joint training and continuation took about 4500, 11,150 and 5574 seconds, respectively; per-plate fine-tuning took about 2700 seconds. The approximately 20-minute model-only timing was measured on an RTX 4500 Ada. We do not infer a controlled runtime speedup from these different machines.

An interpretation amendment was recorded after the initial TV/pretrained-PnP and own-decoder results, before the neuron-trained-PnP and off-design evaluations. Final reporting retains signed differences under every method; it does not classify the coupled plate as unable to transfer. New Fisher/coherence feed-forward evaluations use the same warm-up checkpoint and 3000-step recipe. Additional acquisition conditions are reported only where complete evaluations exist.

S4. Reconstruction and validation

Shared framework. For TV and PnP, the forward operator and validation-selected scalar parameters change across encoders, while denoiser weights and solver families remain fixed. The separately fine-tuned feed-forward route is specified in Sec. S3. TV uses $\frac{1}{2}\|\mathsf{A}\mathbf{x}-\mathbf{y}\|_2^2+\lambda\|\nabla\mathbf{x}\|_{2,1}$ with a box constraint. PnP uses a quadratic data update with denoising and optional sparsity branches. A quadratic data term penalizes residuals but does not guarantee membership in the hard measurement-interval set used in Sec. S9. Generic denoisers need not minimize an explicit regularization energy.

ADMM and plug-and-play updates. Let C_j map the volume to branch j, with auxiliary variable $\mathbf{v}_j$ and scaled dual $\mathbf{d}_j$. Each iteration computes

$$\mathbf{x}^{k+1} = \arg\min_{\mathbf{x}} \Big\{ \tfrac{1}{2}\|\mathsf{A}\mathbf{x}-\mathbf{y}\|_2^2 + \sum_j \tfrac{\rho_j}{2}\|C_j\mathbf{x}-\mathbf{v}_j^k+\mathbf{d}_j^k\|_2^2 \Big\}, \tag{S15}$$

$$\mathbf{v}_j^{k+1} = \mathcal{F}_j(C_j\mathbf{x}^{k+1}+\mathbf{d}_j^k), \tag{S16}$$

$$\mathbf{d}_j^{k+1} = \mathbf{d}_j^k + C_j\mathbf{x}^{k+1} - \mathbf{v}_j^{k+1}. \tag{S17}$$

The data update is solved by conjugate gradients. For TV, C_j is the 3D gradient or identity, and $\mathcal{F}_j$ is vector soft thresholding or box projection. PnP uses identity branch maps and denoisers in place of proximal functions. If P_o rearranges a shared volume into slices of orientation $o \in \{xy, yz, zx\}$, the slice branch is $P_o^{-1}DP_o$, with D acting independently on each slice. Pretrained PnP has these three branches; neuron-trained PnP replaces their shared 2D network and adds a 3D denoising branch. An optional ℓ_1 branch uses scalar soft thresholding. Generic DnCNN branches are not asserted to be proximal maps of a known joint regularizer.

TV. We use isotropic 3D forward-difference TV and ADMM with separate gradient and box branches, both with unit penalty. The box branch projects its auxiliary variable at each iteration, and the final volume is clipped. There are 300 ADMM iterations; each conjugate-gradient volume update uses at most 20 iterations at relative tolerance 10^{-5}. The starting volume is normalized back-projection clipped to $[0, 1]$. The regularization weight is selected from $\{0.01, 0.03, 0.1, 0.3, 1, 3\}$ using mean validation PSNR. Edge selections extend the grid by a factor of three, at most twice. At $C = 8$, validation selects $\lambda = 0.1$ for all five primary designs. These fixed iteration budgets are not claims of exact minimization. Additional checks

with 1000 iterations changed mean PSNR by -0.03 dB at $C = 8$ and centroid error by at most $0.001\,\mu$m; checks at $C = 1$ changed PSNR by at most 0.03 dB and centroid error by $0.006\,\mu$m.

PnP. Both variants run 20 ADMM iterations, with at most 20 conjugate-gradient steps per volume update and relative tolerance 10^{-5}. Denoising-branch penalties are $5s$ and the sparsity penalty is $10s$. The implementation sets nominal branch weights of 0.2 for the slice denoisers and 1.0 for the volumetric denoiser. The fixed denoisers ignore this proximal-scale argument, so these values activate the branches rather than adjusting denoising strength. Their coupling to the data is controlled by the shared penalty. The sparsity weight is $0.2l$; branches with zero weight are omitted. In the primary $C = 8$ comparison, pretrained PnP uses $s = 0.25, l = 0$ for every design. Neuron-trained PnP uses $s = 0.125$ for the untrained plate and $s = 0.25$ for the other designs; $l = 1/16$ only for the seed-0 Fisher plate and zero otherwise. We validate $s \in \{2^{-4}, 2^{-3}, \dots, 4\}$ and $l \in \{0, 1/16\}$. Boundary selections extend the scale grid by a factor of two and the nonzero sparsity scale by four, at most twice. Final selections are interior. All PnP results start from normalized back-projection and are clipped to $[0, 1]$ only after the final iteration.

Denoisers. Pretrained PnP uses SCICO's 17M natural-image DnCNN (17 layers, trained at noise standard deviation 0.1) [2, 14], independently along three slice orientations. Its denoising noise level is fixed; it does not receive a specimen-dependent noise estimate. Neuron-trained PnP replaces this 2D model and adds a 3D branch. The 2D network has 17 layers and 64 channels; the 3D network has 12 layers and 48 channels. We do not evaluate 2D-only and 3D-only neuron-trained variants, so the comparison changes both the training distribution and denoising architecture.

Training uses 459 crops from cell 1, stacks 8 and 9 of animal SOM009, with stack 10 held out for monitoring. Duplicated planes and planes whose mean falls below half their neighbors' median are excluded before crop extraction. Both networks use Adam at 10^{-3}, Gaussian noise $\sigma \sim \mathcal{U}(0, 0.25)$, flips, lateral transpositions and intensity scaling in $[0.3, 1]$. The 2D model uses random slices for 30,000 steps at batch size 64; the 3D model uses volumes for 15,000 steps at batch size 8. Neither contributes test data or is updated during optical training or reconstruction.

Parameter selection. The two vascular validation crops come from another animal and are used for all neuronal comparisons. Parameters are selected separately by encoder, compression and changed noise condition, with the same selection rule and fixed prior weights. No neuronal test crop is used for tuning. Model-only optical design uses neither validation nor test specimens. The supervised counterpart uses its separately held-out neuronal validation volumes as described in Sec. S3. Reference arrays are used only for metric reporting, not in the reconstruction updates.

S5. Data, metrics and paired statistics

Dataset partitions. The neuronal references are public two-photon stacks used in DEEP^2 [12]. Test cells are SOM004 cells 1–2, SOM005 cell 1 and SOM006 cells 1–3: six cells from three animals. Each cell contributes two prespecified crops selected by projected foreground area times depth spread. After fixing reconstruction parameters, eight additional eligible crops per cell were sampled uniformly by location, excluding the two already selected. The resulting 48 crops were reconstructed without retuning. There are 60 test crops in the main comparison; other full-field studies use the 12 prespecified crops unless stated otherwise.

Denoisers and supervised optical/decoder models are trained on SOM009, a fourth animal, as specified in Secs. S3 and S4. TV/PnP parameter selection uses two vascular crops from a different animal; supervised checkpoint selection uses 48 held-out neuronal volumes from SOM009. Conformal calibration uses a held-out stack of SOM009 and does not tune the reconstruction. No test volume enters optical training, denoiser training, scalar tuning or interval calibration.

Profile and centroid errors. Let $g_c = \sum_z x_{\mathrm{gt}}(c, z)$ and retain columns with $g_c > 0.1 \max_{c'} g_{c'}$. We set $\pi_c = g_c / \sum_{c'} g_{c'}$ over retained columns and normalize each nonnegative profile to unit mass. The one-dimensional W1 distance equals Δz times the sum of absolute CDF differences; the main paper averages it with π_c. The implementation clips negative values to zero for the profile metric and divides by column mass plus 10^{-12}. A zero-mass estimate therefore remains a zero profile rather than being assigned a fabricated depth distribution. The centroid error uses the same reference columns and weights, but the feed-forward evaluation computes centroids from raw network outputs; the W1 metric uses their nonnegative part. TV/PnP estimates are clipped before both metrics. For positive, normalized profiles, W1 is at least their absolute centroid difference; this inequality is not asserted for signed feed-forward centroids. A centroid can match when the profile does not, but similar reductions in the two errors do not establish a larger gain in profile shape.

PSNR uses the complete volume and SCICO's default reference range, $\max(\mathbf{x}_{\mathrm{gt}}) - \min(\mathbf{x}_{\mathrm{gt}})$ [2]. Foreground SSIM uses a Gaussian-weighted 3D SSIM map ($\sigma = 1.5$, intensity range one), averaged over reference voxels above 10 % of peak intensity after two binary-dilation steps. These intensity metrics complement W1, which normalizes each column and therefore does not penalize brightness errors by itself.

Depth-blind references. The flat estimate preserves each column's total brightness but distributes it uniformly over depth. The depth-blind oracle preserves that brightness and uses the reference crop's global axial profile for every column. The latter has access to the correct projected structure and global depth distribution but no local depth placement. On the 12 prespecified crops, their W1 errors are 2.266 and $1.432\,\mu\mathrm{m}$, with centroid errors 2.148 and $1.322\,\mu\mathrm{m}$, respectively. They are context for interpreting axial metrics, not reconstruction methods supplied with the actual measurements.

Pairing and uncertainty. For each setting, encoders share the specimen, direction subset, nominal dose and noise-generation protocol. We average the five draws within each crop. A relative change is the ratio of learned and comparator crop means minus one, not the mean of per-crop percentages. Confidence intervals use 10,000 cell-bootstrap resamples, keeping all crops of a sampled cell together. Cells from the same animal may also be dependent; six cells from three animals do not support a precise population-level guarantee. The intervals describe this finite evaluation and are not corrected for all exploratory comparisons.

Consistent reconstruction metrics. W1, centroid, PSNR and foreground SSIM in Tabs. S1, S4 and S6 are computed from the same completed reconstruction rows. Each complete encoder/method configuration contains all 60 crops and five draws. The primary three-plate/four-method table contains 3,600 unique rows. The primary five-design/four-method table contains 6,000 unique rows, including the Fisher and coherence designs. Each criterion configuration has the same 60 crops and five draws. The second optical sampling seed adds 2,700 rows for our, Fisher and coherence plates under TV/PnP, using the same crops. Ancillary 12-crop studies are labeled separately. The reconstruction reruns use the recorded operators, parameters, directions and noise. Relative to earlier runs, the largest per-row differences are $0.019\,\mathrm{dB}/0.0004\,\mu\mathrm{m}$ for TV and $0.198\,\mathrm{dB}/0.0204\,\mu\mathrm{m}$ for PnP (PSNR/centroid); mean metrics agree at the reported precision. The all-column W1 evaluation used a first-row reproduction check for each configuration before completing its remaining solves. We report the resulting numerical estimates rather than assuming bitwise reproducibility across GPU runs.

S6. Full reconstruction results

Three optical designs and four reconstruction methods. Table S4 reports W1, centroid error, PSNR and foreground SSIM from the same 3,600 reconstructions: three plates, four methods, 60 crops and five draws. The feed-forward decoder is fine-tuned separately for each plate, whereas TV and the PnP networks are independent of optical training. Table S5 gives paired cell-bootstrap intervals for W1. Negative changes favor the first named design. The end-to-end plate has a measurable advantage with its own decoder and under both PnP methods, while our plate has lower W1 under TV. The different directions preclude a universal ranking of the two learned plates.

Optical sampling repeats. Table S6 adds our second optical sampling seed to the primary TV/PnP comparison, using the same untrained initialization. Each learned plate lowers W1 on every crop under each method. Seed 0 improves centroid error on 59 of 60 crops for neuron-trained PnP and all crops for the other methods. Seed 1 improves centroid error on all crops. These repeats test sampling variability, not independence of phase initialization.

Reference-based and random crops. Table S7 separates the 12 prespecified crops from the 48 later random crops. All methods retain positive optical gains on the latter, with parameters already fixed. This addresses dependence on the initial structure-score selection, while preserving the cell and animal limitations of the dataset.

Comparison with untrained encoders. The separate centroid study in Tab. S8 uses pretrained PnP, the 12 prespecified crops and the expanded validation grid. Its phase, amplitude and DMD rows share the same $2\times$ aperture, relay, photon model, direction draws and tuning procedure. Learned centroid errors of 0.230–$0.234\,\mu\mathrm{m}$ are below all three random phase plates (0.422–$0.438\,\mu\mathrm{m}$) and amplitude masks (0.396–$0.523\,\mu\mathrm{m}$). The learned seed-0 plate reduces centroid error by 42–56 % against the amplitude masks and 45–47 % against the random phase plates. This shows that its initialization is representative of these controls. It does not show that learning succeeds from other initializations.

The two in-focus DMD pattern sets remain close in centroid error and image quality. Their low depth discrimination concerns this fixed-focus control, not all amplitude encoders. Indeed, the defocused amplitude masks recover depth and can match random phase plates in centroid error. The plate-free scan also retains weak finite-aperture diversity, so its observations are not an ideal zero-depth-information system.

Point-emitter diagnostics. For 25 interior lateral locations and 16 depths, we compare measurement columns under a Gaussian approximation of shot noise with unknown emitter brightness. Random phase plates assign the wrong depth in 1.5–1.7 % of cases with 32 directions; amplitude masks give about 5 %, the in-focus DMD about 25 %, and the plate-free control about 89 %. These are model-based emitter diagnostics, not measured resolution of dense structures.

Reconstruction	Plate	W1 (μm)	Centroid (μm)	PSNR	FG-SSIM
TV	Untrained	0.371	0.259	36.70	0.850
TV	Ours	0.218	0.150	38.29	0.910
TV	End-to-end	0.222	0.151	38.07	0.905
Pretrained PnP	Untrained	0.579	0.490	35.72	0.806
Pretrained PnP	Ours	0.290	0.224	37.56	0.900
Pretrained PnP	End-to-end	0.281	0.212	37.52	0.897
Neuron-trained PnP	Untrained	0.271	0.197	38.01	0.902
Neuron-trained PnP	Ours	0.195	0.139	39.60	0.934
Neuron-trained PnP	End-to-end	0.189	0.131	39.70	0.936
Feed-forward	Untrained	0.366	0.282	36.67	0.872
Feed-forward	Ours	0.265	0.196	38.04	0.912
Feed-forward	End-to-end	0.234	0.167	38.53	0.922

Table S4. Three-plate comparison at $C = 8$ and matched nominal dose, with 60 crops and five draws. Errors are in μm. All metrics in a row use the same reconstructions. Feed-forward networks are fine-tuned for their respective plates; their earlier training histories are specified in Sec. S3.

Reconstruction	Ours vs. untrained	End-to-end vs. ours
TV	-41.3% [-44.5, -37.7]	$+1.9\%$ [$+1.1$, $+2.7$]
Pretrained PnP	-49.9% [-52.8, -46.6]	-3.0% [-5.0, -1.0]
Neuron-trained PnP	-27.9% [-29.5, -26.0]	-3.2% [-5.6, -1.2]
Feed-forward	-27.5% [-31.2, -24.7]	-11.6% [-14.4, -9.0]

Table S5. Paired relative W1 changes (95 % cell-bootstrap intervals) for the same 60-crop comparison. Each change is the ratio of crop-mean errors minus one, after averaging the five draws.

Reconstruction	W1 (μm)	Centroid (μm)	PSNR	FG-SSIM	W1 change [95% CI]
TV	0.218	0.149	38.31	0.910	-41.3% [-44.6, -37.5]
Pretrained PnP	0.289	0.222	37.58	0.900	-50.1% [-53.1, -46.7]
Neuron-trained PnP	0.194	0.136	39.64	0.934	-28.5% [-30.1, -27.1]

Table S6. Our second optical sampling seed at $C = 8$, 60 crops and five draws per crop, with matched nominal dose. Changes compare it with the shared untrained initialization; brackets are cell-bootstrap 95 % intervals. Seed 0 is in Tab. S4.

The plate-free diagnostic uses the original $2\times$ aperture; its neuronal control in Tab. S8 uses $4\times$.

Comparisons across priors. A descriptive pairing of learned plate + TV against random plate + neuron-trained PnP gives 19.6 % lower W1 (95 % CI: 16.4–22.5 %), improving 56 of 60 crops. It does not establish a general dominance relation: learned plate + pretrained PnP has 7 % higher mean W1 than random plate + neuron-trained PnP (interval -2.6 to $+17.3$ %). Optical and prior choices interact; the comparisons in this paragraph concern only the untrained plate and our design, not the end-to-end plate.

S7. Structural tasks and measurement discrimination

Crossing phantoms. We build Gaussian fluorescent tubes on the $32 \times 96 \times 96$ analysis grid, at $0.32\,\mu$m spacing. Their lateral and axial standard deviations are $0.32\,\mu$m, with half-length $12.8\,\mu$m. The equal-branch reconstruction phantoms combine tubes by their maximum before $2 \times 2 \times 2$ averaging. Three lateral geometries use filament-a orientation/crossing-angle pairs $(0°, 90°)$, $(30°, 60°)$ and $(60°, 45°)$. Centers are reconstruction planes 7 and 5. Equal-branch gaps are 0, 2, 3, 4, 6 and 8 planes (0, 1.28, 1.92, 2.56, 3.84 and 5.12 μm), giving 36 phantoms, each with five measurement draws. All use $C = 8$, matched nominal dose and parameters selected on the vascular validation crops, not on these phantoms.

Reconstruction	Seed	Crops	W1 change [95 % CI]	W1 improved	Centroid change [95 % CI]
TV	0	12 prespecified	−38 % [−43, −34]	12/12	−37 % [−41, −34]
TV	0	48 random	−42 % [−46, −37]	48/48	−43 % [−49, −38]
TV	1	12 prespecified	−39 % [−42, −36]	12/12	−38 % [−40, −36]
TV	1	48 random	−42 % [−46, −37]	48/48	−44 % [−49, −38]
Pretrained PnP	0	12 prespecified	−44 % [−50, −39]	12/12	−46 % [−53, −41]
Pretrained PnP	0	48 random	−51 % [−54, −47]	48/48	−56 % [−59, −52]
Pretrained PnP	1	12 prespecified	−44 % [−49, −38]	12/12	−46 % [−52, −40]
Pretrained PnP	1	48 random	−51 % [−55, −47]	48/48	−57 % [−60, −52]
Neuron-trained PnP	0	12 prespecified	−27 % [−34, −20]	12/12	−26 % [−35, −18]
Neuron-trained PnP	0	48 random	−28 % [−30, −26]	48/48	−31 % [−34, −28]
Neuron-trained PnP	1	12 prespecified	−29 % [−34, −25]	12/12	−29 % [−35, −24]
Neuron-trained PnP	1	48 random	−28 % [−30, −27]	48/48	−31 % [−34, −29]

Table S7. Separate evaluation on the prespecified and randomly sampled crops. Negative changes favor the learned plate. Both sets contain crops from the same six cells; the 48 additional crops expand spatial coverage rather than biological replication.

Encoder	Centroid (μm)	PSNR (dB)	FG-SSIM
Learned, seed 0	0.230	36.15	0.868
Learned, seed 1	0.234	36.21	0.869
Random phase 0	0.431	34.81	0.784
Random phase 1	0.422	34.93	0.787
Random phase 2	0.438	34.85	0.782
Amplitude mask 0	0.523	33.70	0.703
Amplitude mask 1	0.419	34.37	0.766
Amplitude mask 2	0.396	34.36	0.763
DMD set 0	2.169	28.83	0.482
DMD set 1	2.169	28.81	0.479
Plate-free, 4× aperture	1.102	30.18	0.494

Table S8. Untrained-encoder control study at $C = 8$: 12 prespecified crops, pretrained PnP. W1 was not computed in this study. The plate-free row uses a 4× aperture in a separate study; its own random-plate control gives 0.446 μm/34.82 dB. Other rows use 2× apertures. Do not interpret the plate-free row as an exactly matched cross-aperture contrast.

Resolved peaks and arm ordering. Peaks are detected on zero-padded axial columns, requiring height and prominence of at least 20 % of that column's maximum. Reference and reconstructed peaks are matched one-to-one within one reconstruction plane, closest first. Crossing columns must carry more than 20 % of each reference filament's maximum column signal and contain reference peaks matched to both planes. The resolved fraction is the share of reference crossing-column signal in columns whose reconstruction has both matched peaks. It is not an absolute recovered-brightness cutoff.

For ordering, each filament's arm columns carry at least 20 % of its own maximum column signal and less than 2 % of the other filament's. Their reconstructed signal-weighted axial centroids must follow the true order, with consecutive arms separated by more than half a plane. A volume succeeds when the resolved fraction is at least 0.5 and ordering is correct. In the zero-gap control, two matched peaks instead count as a false split.

The qualifying gap is the smallest tested gap of at least two planes with success rate at least 0.8 at that gap and every larger tested gap, at both centers separately. Each center's rate uses 15 reconstructions (three geometries and five draws). The main figure pools the two centers in each curve; the criterion uses their individual rates.

The displayed example uses the first lateral geometry, center 7 and draw 0. Its gap was chosen at our plate's TV threshold, so the gap is outcome-informed. Untrained and learned TV succeed in 10 % and 90 % of the pooled cases at that gap; the displayed draw is one failed/successful pair. All displayed sections share geometry and intensity scale. Neither plate falsely splits the merged control.

Weak and three-filament extensions. The weak-branch cases use the same two-filament geometries with the deeper branch at 0.3 times the first's intensity. Gaps are 0, 3, 4, 6 and 8 planes; at two planes the reference itself fails the peak criterion. Only TV with our plate meets the two-center

	Equal branches		Weak branch		Three filaments	
Reconstruction	Untrained	Ours	Untrained	Ours	Untrained	Ours
TV	3.84	2.56	–	3.84	–	3.20
Pretrained PnP	3.84	2.56	–	–	–	3.20
Neuron-trained PnP	3.84	3.84	–	–	–	–

Table S9. Qualifying gaps (μm) under the same success rule. A dash means the criterion is not reached at both centers over the tested gaps. Weak branch: the second filament has 0.3 times the first's intensity. The three-filament columns include the descriptive wider-gap extension: $3.20\,\mu$m was tested at both centers, and $3.84\,\mu$m only at center 7, since the other geometry would leave the field.

threshold, at $3.84\,\mu$m (Tab. S9).

Three equal filaments occupy planes $c-d, c, c+d$, with orientations $(0°, 60°, 120°)$ or $(30°, 90°, 150°)$. Initially tested spacings $d = 0, 2, 3, 4$ planes do not achieve 80 % success. A descriptive extension adds five planes at both centers and six at center 7, using the same parameters and success rule. Our plate then resolves all three filaments at five-plane spacing ($3.20\,\mu$m) under TV and pretrained PnP at both centers. The untrained plate reaches 80 % only at center 7: six planes under TV and five under pretrained PnP. With neuron-trained PnP, our plate reaches the criterion at six planes at center 7 but not at center 5; the untrained plate reaches it at neither center. These extensions show the dependence on spacing and prior, rather than a general guarantee for denser structures.

Joint-training control. The end-to-end plate is evaluated on the same equal-branch phantoms with TV and pretrained PnP. Pooling the five positive gaps, its success rate minus ours is -3.3 percentage points under TV (95 % paired phantom-bootstrap interval: -8.7 to $+1.3$) and $+0.7$ under pretrained PnP (-1.3 to $+2.7$). Thus, full-dose success differences between the learned plates are not established. Reduced-dose comparisons are given in Sec. S8.

Merged neuronal alternatives. A reference column is called layered when it belongs to the signal mask and has at least two peaks under the same relative height/prominence rule. For each corresponding fine-grid lateral column, we move its intensity to its axial centroid, distributing mass between the two neighboring planes to preserve that centroid exactly. Other columns remain unchanged. The resulting alternative preserves the fine-grid lateral projection and each modified column's centroid while changing its depth profile. One crop has no eligible columns, leaving 59. These are constructed candidates, not solutions optimized over a measurement-consistency set; concentrated intensities need not remain below the original normalized peak.

For noise-free expected counts μ of the reference and $\nu(c)$ of a brightness-scaled alternative, we compute

$$D_{\mathrm{P}}(\mu\|\nu) = \sum_i \left[\mu_i \log \frac{\mu_i}{\nu_i} + \nu_i - \mu_i\right]. \tag{S18}$$

The simulator evaluates both objects on the fine grid, using each matched direction subset and the same exposure. Without background, the profiled scale is the ratio of total reference to alternative expected counts. With background, the numerical fit searches a log-scale interval of $[-1, 1]$ around the total-signal estimate. Crop scores average five subsets. The geometric mean of the crop-level learned/untrained ratios is 9.99 (95 % CI: 8.66–11.79) without background and 10.4 with background. Against the end-to-end plate, our ratio is 2.00 (1.91–2.09), with higher divergence on all 59 eligible crops. The draw-average is taken within each crop before forming the ratios. Random-plate divergences can already be large; the ratio is a diagnostic of these specified alternatives, not a probability of successful reconstruction. No KL-to-reconstruction-success guarantee is claimed.

Additional neuronal structure metrics. Layered columns carry about 17 % of reference signal on average. Their W1 decreases by 33.8 % with TV, 42.4 % with pretrained PnP and 27.6 % with neuron-trained PnP. However, the prespecified comparison of profile gain against centroid gain does not support a larger profile benefit: for TV the difference is -0.7 percentage points (95 % CI: -2.9 to $+1.5$). Most layered columns are intensity structure within thick branches, so these metrics alone do not establish separated neuronal branches.

The earlier layered-column protocol required each rerun to reproduce its recorded PSNR within 0.05 dB. TV met that check; PnP did not. The PnP layered-column results are therefore descriptive. The later all-column W1 evaluation uses its separately documented numerical check (Sec. S5). We do not relabel the failed earlier gate as a passed structural test.

As a further descriptive measure, centerline Dice [11] compares a reference mask at 10 % of peak intensity with an estimated mask retaining the same number of voxels, followed by 3D skeletonization. Scores rise from 0.79 to 0.84

for TV, 0.75 to 0.84 for pretrained PnP and 0.85 to 0.88 for neuron-trained PnP. This thresholded metric is depth-sensitive but does not substitute for branch tracing or connectivity validation.

S8. Measurement, photon and alignment controls

Dose convention. Each encoder's excitation is divided by its mean over the specimen analysis grid and the full 256-direction scan. This fixes the deposited excitation per complete scan. For k directions, exposure is multiplied by $256/k$. Random subsets therefore match nominal dose in expectation, not exactly for each subset. The per-specimen exposure is set from the random plate's full-scan mean signal and reused for every encoder. Equal deposited dose does not force identical detected counts: illumination can correlate differently with the specimen. For our two optical sampling seeds, mean detected signal changes by about -1 to $-2\,\%$ across the 12 crops, with individual changes approximately -12 to $+9\,\%$.

Measurement compression. Table S10 reports both depth metrics at $C = 1, 4, 8$. These values count all camera samples, not camera images alone. At matched nominal dose, fewer directions mean longer exposure per image. The learned 32-image acquisition gives lower W1 than the 256-image random acquisition for TV and pretrained PnP, but the close neuron-trained-PnP means do not establish equivalence. The TV curve is not monotone in compression: validation can prefer a stronger regularization weight at $C = 4$, which smooths neuronal features more than at $C = 8$.

Noise conditions. The nominal model uses Poisson shot noise. The background condition adds a uniform mean equal to the random plate's mean expected signal for the same exposure, plus independent Gaussian read noise with standard deviation 2 electrons. The known mean background is subtracted before reconstruction. The same background level, exposure and noise protocol are used for each encoder. A separate low-dose condition reduces exposure to one tenth. Scalar reconstruction parameters are revalidated for each changed noise condition, while networks stay fixed. Table S11 reports all three methods.

Reduced-dose joint-training comparison. At one tenth of the nominal dose, both learned plates are evaluated with TV and pretrained PnP on the 12 prespecified neuronal crops and on equal-branch crossings. Networks remain fixed and scalar parameters follow the condition-specific validation protocol. Neuronal W1 differences are not established (Tab. S12a). In the crossing task, however, our plate succeeds more often when pooling the three positive gaps 1.28, 1.92 and 2.56 μm (Tab. S12b and Fig. S1). Each gap has six phantom cases (three geometries, two centers), with three draws per case. The paired bootstrap averages draws and resamples the 18 cases, rather than treating draws as independent phantoms. This comparison is a targeted structure stress test, not evidence of generally better low-dose neuronal reconstruction.

Photon budget. The dose sweep uses factors 0.1, 0.3, 1 and 3 of the nominal exposure (Tab. S13). At dose 0.3, learned-plate W1 is 0.234, 0.302 and 0.220 μm for TV, pretrained PnP and neuron-trained PnP, compared with 0.318, 0.496 and 0.282 μm for the random plate at dose 1. These reductions of 26 %, 39 % and 22 % describe the tested simulation and inference budgets. The random curve's flattening between doses 1 and 3 is not evidence of a fundamental photon limit.

Alignment. Distance perturbations set $D = 6.1$ or $7.1\ \mu$m. In the characterized case, the operator is rebuilt from the known distance and scalar reconstruction parameters are revalidated; the phase is unchanged. In the nominal-operator case, reconstruction ignores the perturbation. Lateral error shifts the plate by one 0.32 μm feature. Pointing errors are offsets of 0.005 or 0.01 in transverse direction cosine. All comparisons use neuron-trained PnP, $C = 8$, the same 12 crops and five draws.

Table S14 separates these assumptions. Known distance changes retain the optical gain. Ignoring distance or lateral registration can make the learned plate worse than the random one, including an 11 % W1 increase for a one-feature lateral shift. Rebuilding an operator with a known perturbation is not an experimentally demonstrated calibration procedure. Phase-quantization experiments are not used to claim a fabricable low-level plate: the simulated surround was not quantized with the trained center.

S9. Complementary uncertainty analyses

These analyses ask different questions from the main reconstruction tests. Centroid intervals quantify an empirical depth error rate after reconstruction. Set-membership bounds assess admissible voxel intensities without a learned prior. Neither supplies a connectivity guarantee for the neuronal reconstructions.

Conformal centroid intervals. We form $\hat{z}_c \pm \lambda$ around each reconstructed axial centroid. For a volume, $L(\lambda)$ is the reference-signal-weighted fraction of evaluated columns whose true centroid lies outside the interval. It is bounded by one and decreases with λ. Conformal risk control [1] selects

$$\hat{\lambda} = \inf\left\{\lambda : \frac{n}{n+1}\hat{R}_n(\lambda) + \frac{1}{n+1} \le \alpha\right\}, \qquad \text{(S19)}$$

Reconstruction	Plate	W1, $C = 1$	Centroid	W1, $C = 4$	Centroid	W1, $C = 8$	Centroid
TV	Random	0.249	0.187	0.298	0.223	0.318	0.243
TV	Learned, seed 0	0.170	0.135	0.208	0.166	0.196	0.153
TV	Learned, seed 1	0.160	0.124	0.202	0.158	0.195	0.151
Pretrained PnP	Random	0.413	0.357	0.445	0.384	0.496	0.431
Pretrained PnP	Learned, seed 0	0.222	0.184	0.254	0.212	0.277	0.231
Pretrained PnP	Learned, seed 1	0.213	0.173	0.249	0.206	0.279	0.234
Neuron-trained PnP	Random	0.203	0.161	0.234	0.184	0.282	0.221
Neuron-trained PnP	Learned, seed 0	0.142	0.110	0.167	0.130	0.206	0.163

Table S10. Measurement budgets on the 12 prespecified crops. W1 and centroid errors are in μm and come from the same reconstruction rows. $C = 1$ uses all 256 directions; $C = 4, 8$ average five subsets of 64 and 32 directions. The second seed is available across these budgets for TV and pretrained PnP; neuron-trained PnP is evaluated with seed 0 here and both seeds at $C = 8$.

Condition	Reconstruction	W1, random	W1, learned	Centroid, random	Centroid, learned	W1 change	Centroid change
Shot noise	TV	0.318	0.196	0.243	0.153	−38 % [−43, −34]	−37 % [−41, −34]
Shot noise	Pretrained PnP	0.496	0.277	0.431	0.231	−44 % [−50, −39]	−46 % [−53, −41]
Shot noise	Neuron-trained PnP	0.282	0.206	0.221	0.163	−27 % [−34, −20]	−26 % [−35, −18]
Background + read noise	TV	0.324	0.227	0.249	0.180	−30 % [−38, −22]	−28 % [−36, −20]
Background + read noise	Pretrained PnP	0.627	0.297	0.548	0.247	−53 % [−57, −48]	−55 % [−60, −50]
Background + read noise	Neuron-trained PnP	0.288	0.210	0.228	0.166	−27 % [−35, −19]	−27 % [−36, −18]
0.1 dose	TV	0.466	0.292	0.365	0.236	−37 % [−45, −29]	−35 % [−44, −28]
0.1 dose	Pretrained PnP	0.834	0.407	0.733	0.352	−51 % [−57, −45]	−52 % [−58, −45]
0.1 dose	Neuron-trained PnP	0.345	0.239	0.269	0.186	−31 % [−37, −24]	−31 % [−38, −24]

Table S11. Noise controls on 12 crops at $C = 8$. Errors are in μm; changes and cell-bootstrap 95 % intervals compare the learned and random plates within the same condition.

(a) Neuronal axial-profile W1 (μm), 12 crops

Reconstruction	Ours	End-to-end	Difference [95% CI]
TV	0.292	0.285	−0.007 [−0.028, +0.011]
Pretrained PnP	0.407	0.405	−0.001 [−0.019, +0.015]

(b) Crossing success, 18 phantom cases

Reconstruction	Ours (%)	End-to-end (%)	Difference (points)
TV	16.7	0.0	−16.7 [−29.6, −5.6]
Pretrained PnP	38.9	24.1	−14.8 [−29.6, −1.9]

Table S12. Joint-training control at $0.1\times$ nominal dose and $C = 8$. Differences are end-to-end minus ours. Intervals resample cells for neuronal W1 and phantom cases for crossing success, after averaging measurement draws. Negative success differences favor our plate; negative W1 differences favor end-to-end.

where $\hat{R}_n$ averages the loss over n calibration volumes. Under exchangeability of the calibration volumes and a new volume, $\mathbb{E}[L(\hat{\lambda})] \leq \alpha$. This controls expected signal-weighted miscoverage, not each column's or specimen's coverage.

We use neuron-trained PnP at $C = 8$, target $\alpha = 0.1$ and encoder-specific calibration on 35 non-overlapping crops from a held-out SOM009 stack. A 240-crop overlapping set is a sensitivity analysis, not independent replication. On the 12 prespecified test crops, the learned seed-0 half-width decreases from 0.646 to 0.474 μm (26.7 %), with empirical miscoverage 0.075 versus 0.060 for the random plate (Tab. S15). The test animals differ from the calibration animal, so exchangeability is not established; observed miscoverage is empirical rather than a cross-animal guarantee.

Prespecified connectivity calls based on reconstructed bottlenecks did not meet their error target: false-disconnected rates were 0.23–0.48 against a 0.15 limit, with no connected test call. We therefore do not convert centroid calibration into error-controlled connectivity statements.

Voxel bounds without a learned prior. Given $\mathbf{x} \in [0, 1]^N$, define

$$\mathcal{C}(\mathbf{y}) = \{\mathbf{x} : \ \ell_i \leq (\mathsf{A}\mathbf{x})_i \leq h_i, \ 0 \leq \mathbf{x} \leq 1\}. \qquad \text{(S20)}$$

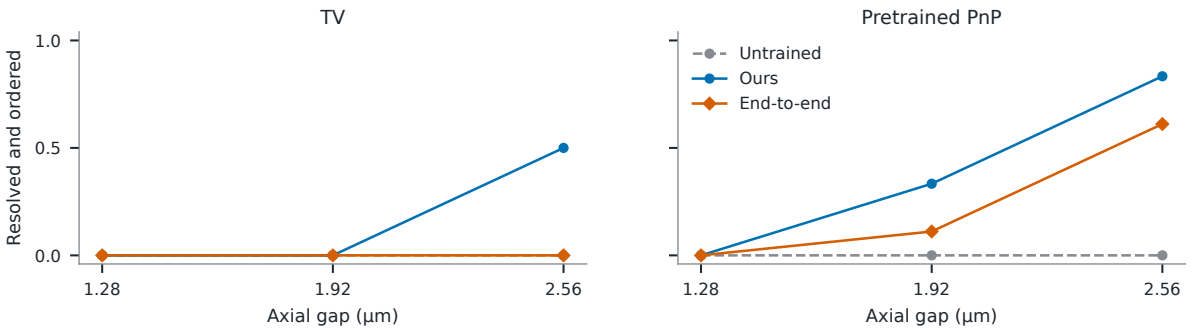


Figure S1. Full low-dose crossing curves for the tested positive gaps. Each point pools the two axial centers, three lateral geometries and three draws. At zero gap, none of the tested plate–method combinations falsely splits the merged structure.

Reconstruction	Nominal dose	W1, random	W1, learned	Centroid, random	Centroid, learned
TV	0.1	0.466	0.292	0.365	0.236
TV	0.3	0.442	0.234	0.341	0.185
TV	1	0.318	0.196	0.243	0.153
TV	3	0.315	0.192	0.242	0.151
Pretrained PnP	0.1	0.834	0.407	0.733	0.352
Pretrained PnP	0.3	0.631	0.302	0.550	0.251
Pretrained PnP	1	0.496	0.277	0.431	0.230
Pretrained PnP	3	0.494	0.276	0.430	0.230
Neuron-trained PnP	0.1	0.345	0.239	0.269	0.186
Neuron-trained PnP	0.3	0.304	0.220	0.236	0.174
Neuron-trained PnP	1	0.282	0.207	0.221	0.164
Neuron-trained PnP	3	0.275	0.203	0.218	0.162

Table S13. Dose sweep at $C = 8$, 12 crops. W1 and centroid errors (μm) are computed from the same reconstructions. Each exposure factor is relative to the matched nominal-dose protocol.

Condition	W1, random	W1, learned	Centroid, random	Centroid, learned	W1 change	Centroid change
none (design)	0.282	0.206	0.221	0.163	−27 % [−34, −20]	−26 % [−35, −18]
Known: D=6.1	0.285	0.205	0.225	0.162	−28 % [−35, −22]	−28 % [−37, −20]
Known: D=7.1	0.284	0.209	0.226	0.166	−27 % [−34, −20]	−26 % [−35, −19]
Nominal operator: scan tilt 0.005	0.318	0.254	0.253	0.196	−20 % [−25, −16]	−22 % [−30, −15]
Nominal operator: scan tilt 0.01	0.379	0.328	0.308	0.251	−13 % [−16, −10]	−18 % [−22, −15]
Nominal operator: D=6.1	0.563	0.586	0.528	0.571	+4 % [+0.3, +8]	+8 % [+3, +14]
Nominal operator: D=7.1	0.447	0.473	0.383	0.438	+6 % [+5, +7]	+14 % [+12, +17]
Nominal operator: lateral shift 1 feature	0.451	0.500	0.372	0.400	+11 % [+2, +23]	+7 % [−0.4, +19]

Table S14. Alignment controls with neuron-trained PnP. Errors are in μm; changes compare learned and random plates under the same perturbation. “Known” means the offset is supplied to operator assembly and validation; “nominal operator” means it is ignored. Positive changes favor the random plate.

For a nonempty set, minimizing and maximizing x_j gives the smallest and largest voxel intensities compatible with the stated constraints [5, 8]. We solve these linear programs with HiGHS [6]. The reconstruction’s quadratic data term does not automatically place its output in this set.

The reduced field contains $4 \times 12 \times 12$ reconstruction voxels and 3×3 camera samples per image, using 32 fixed directions. Thus, $N = 576$ and $M = 288$; this is a separate geometry, not the full-field $C = 8$ experiment. Plates are trained separately on this reduced geometry; no reflected-depth pair qualifies for the full-grid weighting rule, so all off-diagonal weights are one.

The ancillary Gaussian model uses reference full-scan mean m_p for the random plate and mean m_e for an encoder, with $\sigma_e = m_p 10^{-q/10}\sqrt{k/256}\sqrt{m_e/m_p}$ and reference parameter $q = 15$. At $q = 15$, this is a spatially uniform Gaussian approximation to the main photon model at each encoder’s mean signal, rather than a pixel-

	35 non-overlapping crops		all 240 crops	
plate	$\hat{\lambda}$ (μm)	test miscov.	$\hat{\lambda}$ (μm)	test miscov.
random	0.646	0.060	0.570	0.085
learned, seed 0	0.474	0.075	0.422	0.091
learned, seed 1	0.486	0.060	0.422	0.081

Table S15. Calibrated centroid half-widths and empirical signal-weighted test miscoverage. The 240-crop analysis reuses overlapping regions. Seed 1 gives $0.486\,\mu$m on the primary 35-crop calibration set.

wise Poisson variance. The formula defines q explicitly; it is not a standard RMS SNR. Intervals use $y_i \pm z\sigma_e$, where $z = \Phi^{-1}(1 - 0.05/(2 \cdot 2880))$. The family contains all 2304 rows of the 256-direction reduced scan plus up to 576 projection constraints. Using the same correction for every subset gives at least 95 % simultaneous coverage under the stated Gaussian model and exact reduced linear operator. Unmodeled optical or specimen discretization error would require a separate allowance.

Across 24 synthetic phantoms and three noise realizations each, the learned reduced-field plates narrow mean voxel-wise bound width by about 21 %. On 48 neuronal windows, however, bounds span the full admissible range for every voxel at the tested settings. Better neuronal reconstruction in the main study therefore does not imply unique voxel intensities or uniquely identified connections. These negative neuronal results are retained as limits of the measurement-only analysis, not as optical-learning performance measures.

S10. Neuronal views and display selection

Reference-only selection. Before opening reconstructions for figure selection, we computed reference features for all 60 crops. Eligible crops had at least 10 skeleton endpoints (the upper quartile) and a 10–90 % column-centroid range of at least $3.64\,\mu$m (the median), indicating several branches and substantial depth extent. Eight crops qualified. We ranked them by reference signal in columns with at least two axial peaks. The first was SOM004_cell2_x7; the companion was the highest-ranked eligible crop from another animal, SOM006_cell1_x7.

After selection, their W1-gain ranks among 60 crops were 52/45/23 and 45/46/43 for TV/pretrained/neuron-trained PnP, where rank 1 is the largest reduction. These are typical-to-lower-gain examples. The main neuronal figure displays pretrained PnP, the prior with the largest cohort-average gain; Fig. S2 shows all priors for that same crop. No larger reconstructed field exists in this evaluation.

Views and scales. The axial band maximizes reference signal in layered columns over eight consecutive lateral rows ($5.12\,\mu$m). Its lateral window spans 24 voxels ($15.36\,\mu$m), centered on that reference signal. Lateral context is the full maximum-intensity projection; axial views are maxima over the selected band with true physical aspect ratio and depth increasing downward. Linear display limits are zero and the reference projection's 99.5th percentile, shared across the matched panels of each row. There is no gamma adjustment, smoothing or per-panel rescaling. Error maps use the reference signal mask and a common 0–2 μm W1 scale. All displayed volumes use draw 0.

A feature missed by both plates. We select one column in the companion band using reference signal alone: the layered column with largest reference mass whose two highest peaks are at least three planes apart. At $(x, y) = (18, 3)$, a weak shallow peak at plane 2 lies above a deeper peak at plane 9. The shallow peak is 24.5 % of the column maximum and carries 7.8 % of its signal in planes 1–3. None of the six reconstructions has a matched shallow peak; that signal becomes a low plateau (Fig. S6). This limit coexists with lower average W1 and improved placement of brighter features.

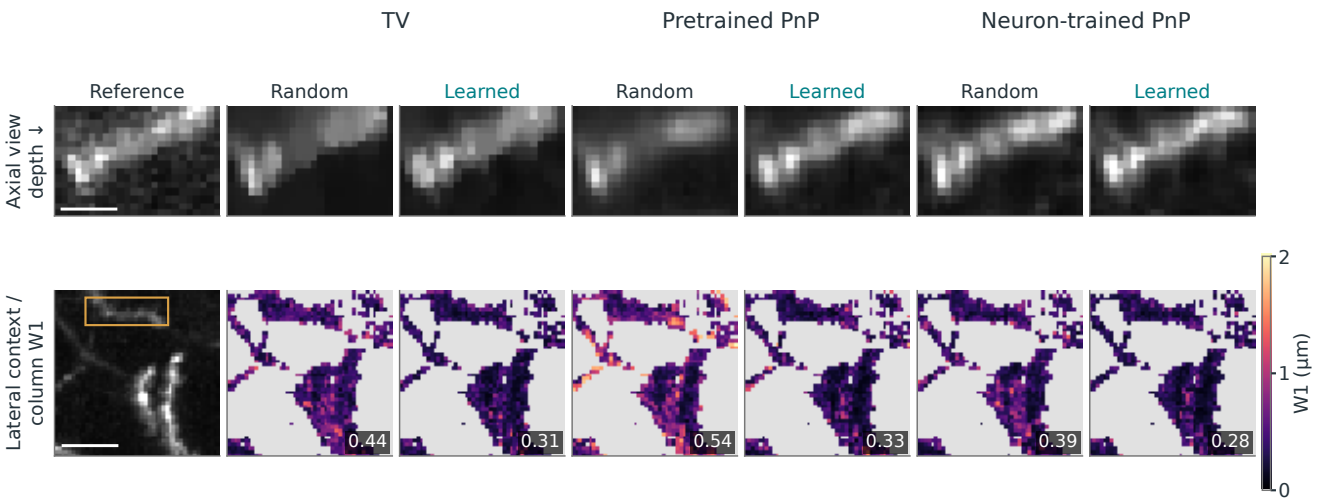


Figure S2. All three priors for the first reference-selected crop in the main neuronal figure. Top: matched axial band views; bottom: reference lateral context and column-W1 maps. Gray is outside the reference signal mask. Panel values are signal-weighted crop W1 for this draw, in μm. Scale bars: 5 μm in band views and 10 μm in lateral context.

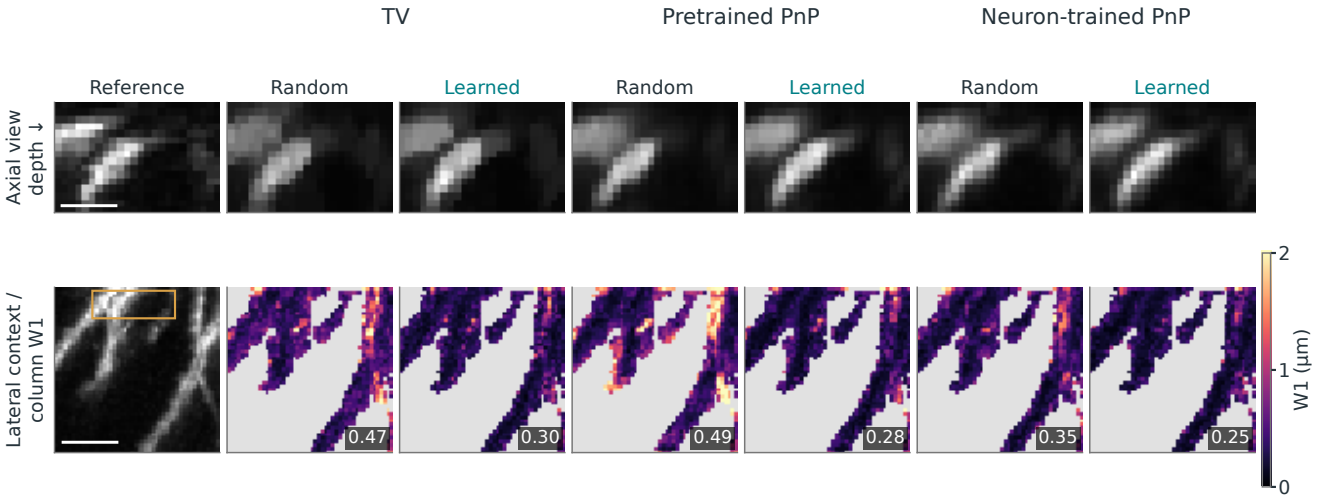


Figure S3. The reference-selected companion from another animal. Top: matched axial band views; bottom: reference lateral context and column-wise W1 maps. Panel values are draw-0 signal-weighted crop W1 in μm. Shared linear scales and view conventions match the first crop. Bars: 5 μm in band views and 10 μm in lateral context.

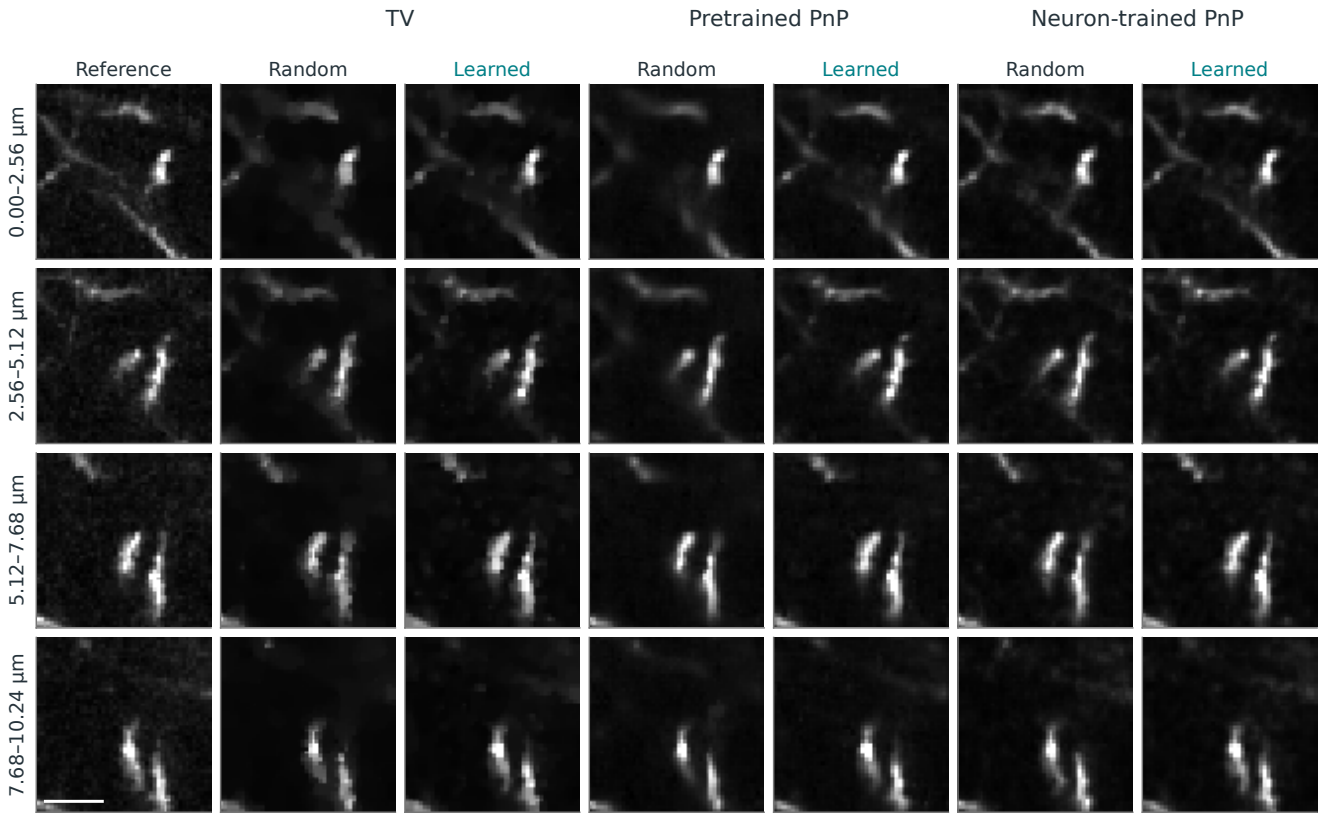


Figure S4. Depth slabs of the first neuronal example, with maximum projections over four consecutive depth planes ($2.56\,\mu$m per slab). Reference and all reconstructions share the same linear intensity range across slabs. Slab views expose signal spread into depths absent from the corresponding reference slab. These are qualitative placement comparisons, not branch-separation measurements. Bar: $10\,\mu$m.

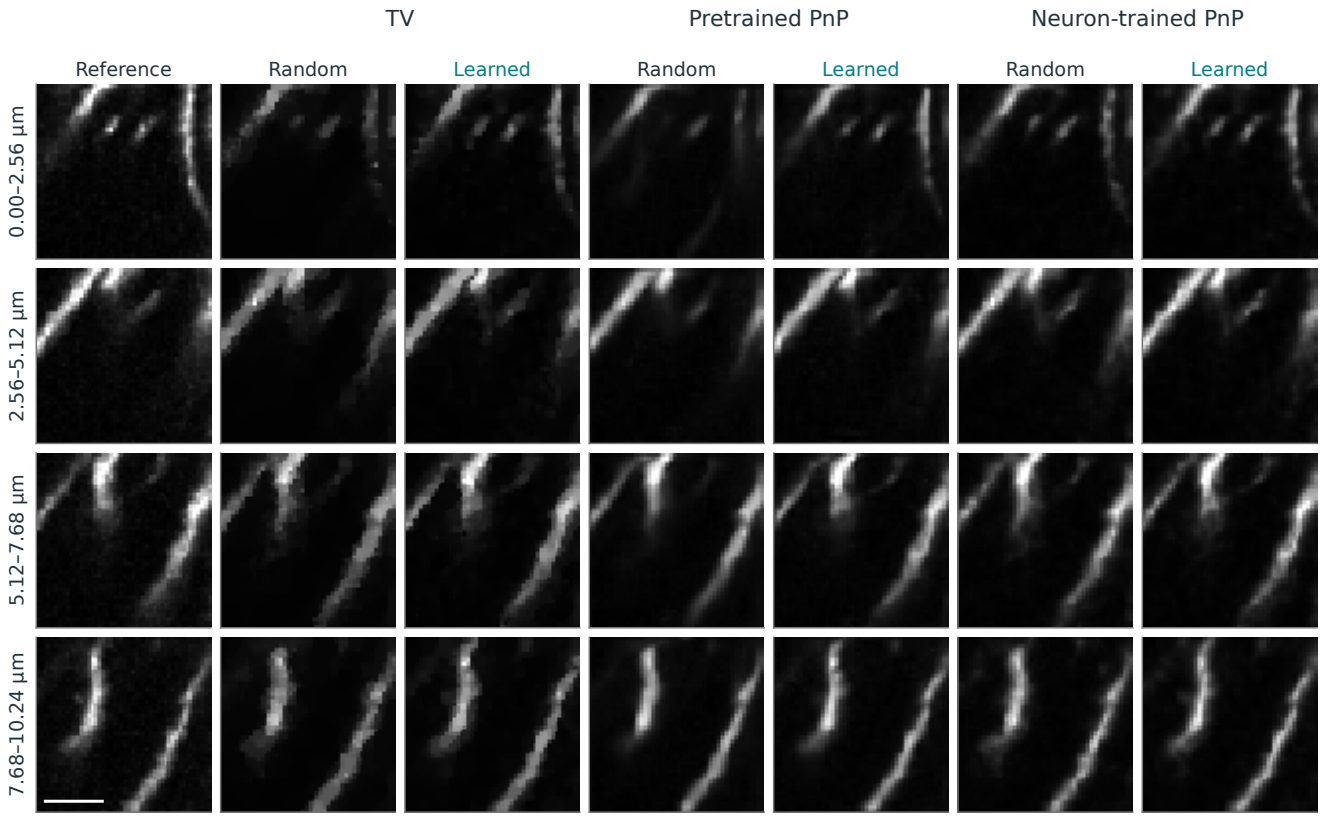


Figure S5. The companion crop from another animal, with the same depth-slab conventions. Stronger priors reduce diffuse signal for both plates; the optical difference is smaller under neuron-trained PnP. The full views retain features missed by either encoder.

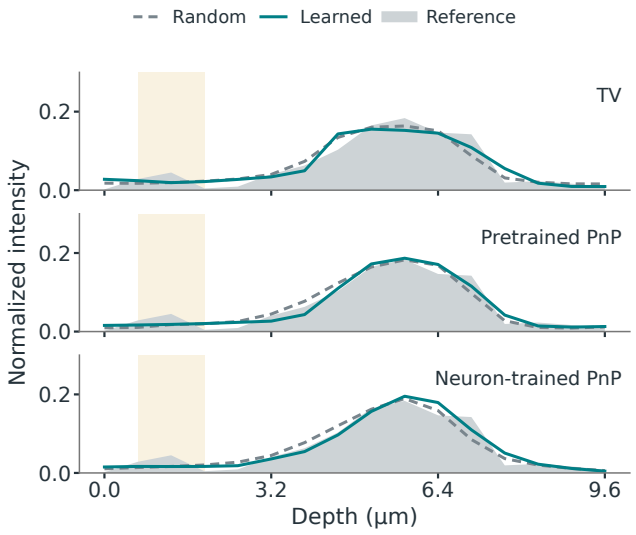


Figure S6. Reference-selected weak-feature column in the companion crop. Profiles are normalized to unit mass. Both plates miss the shallow reference peak under all three priors, despite the lower average W1 of the learned plate. This column was fixed from reference structure before inspecting reconstructions.

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